semmcci: Internal Tests

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Tests

```
#> test-semmcci-mc-func-simple-med
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-semmcci-mc-generic-simple-med-defined

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 1 success.

#> test-semmcci-mc-latent-med-defined-none

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-semmcci-mc-latent-med-defined

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-semmcci-mc-latent-med-std-defined-none

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-semmcci-mc-latent-med-std-defined

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-semmcci-mc-moment
```

```
#> Test passed with 1 success.
```

#> test-semmcci-mc-simple-med-defined-equality

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-defined-inequality

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-defined-meanstructure

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-defined-mi-amelia

```
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-defined-mi-list

```
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-defined-mi-mice

```
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-defined-none

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-defined

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-std-defined-meanstructure

```
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-semmcci-mc-simple-med-std-defined-none-random-x

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-std-defined-none

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-simple-med-std-defined

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

#> test-semmcci-mc-methods

```
#> Monte Carlo Confidence Intervals
#>                     est     se   R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> visual=~x1       1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> visual=~x2       0.5535 0.0956 100 0.3054 0.3161 0.3785 0.7578 0.7774 0.7793
#> visual=~x3       0.7294 0.1060 100 0.4822 0.4980 0.5187 0.9234 0.9375 0.9420
#> textual=~x4      1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> textual=~x5      1.1131 0.0691 100 0.9474 0.9558 0.9695 1.2482 1.2733 1.2857
#> textual=~x6      0.9261 0.0582 100 0.7721 0.7912 0.8281 1.0546 1.0777 1.0816
#> speed=~x7        1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> speed=~x8        1.1800 0.1688 100 0.7850 0.7918 0.8702 1.5590 1.6317 1.6502
#> speed=~x9        1.0815 0.1491 100 0.7197 0.7266 0.7852 1.3430 1.4521 1.4920
#> x1~~x1           0.5491 0.1157 100 0.2287 0.2449 0.3644 0.8219 0.9108 0.9177
#> x2~~x2           1.1338 0.1084 100 0.8091 0.8654 0.9549 1.3546 1.3900 1.4003
#> x3~~x3           0.8443 0.0929 100 0.6397 0.6413 0.6650 1.0199 1.0315 1.0317
#> x4~~x4           0.3712 0.0473 100 0.2398 0.2471 0.2809 0.4553 0.4655 0.4692
#> x5~~x5           0.4463 0.0627 100 0.3034 0.3099 0.3458 0.5672 0.6153 0.6427
#> x6~~x6           0.3562 0.0447 100 0.2676 0.2691 0.2776 0.4474 0.4608 0.4643
#> x7~~x7           0.7994 0.0718 100 0.6239 0.6334 0.6664 0.9397 0.9512 0.9528
#> x8~~x8           0.4877 0.0757 100 0.3144 0.3235 0.3429 0.6331 0.6621 0.6742
#> x9~~x9           0.5661 0.0636 100 0.3842 0.4105 0.4425 0.6850 0.7266 0.7402
#> visual~~visual   0.8093 0.1267 100 0.4800 0.5015 0.5613 1.0474 1.0915 1.1059
#> textual~~textual 0.9795 0.1161 100 0.6970 0.7120 0.7434 1.1894 1.2546 1.2780
#> speed~~speed     0.3837 0.0847 100 0.1291 0.1454 0.2122 0.5275 0.5479 0.5597
#> visual~~textual  0.4082 0.0678 100 0.2315 0.2322 0.2566 0.5179 0.5383 0.5467
#> visual~~speed    0.2622 0.0505 100 0.1306 0.1354 0.1603 0.3536 0.3744 0.3756
#> textual~~speed   0.1735 0.0482 100 0.0696 0.0706 0.0805 0.2706 0.2856 0.2909
#> Monte Carlo Confidence Intervals (Multiple Imputation Estimates)
#>                     est     se   R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> visual=~x1       1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> visual=~x2       0.5535 0.1011 100 0.2572 0.2843 0.3692 0.7636 0.8380 0.8663
```

```
#> visual=~x3       0.7294 0.1203 100 0.2924 0.3508 0.4939 0.9330 0.9790 0.9896
#> textual=~x4      1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> textual=~x5      1.1131 0.0573 100 0.9818 0.9891 1.0032 1.2323 1.2600 1.2649
#> textual=~x6      0.9261 0.0538 100 0.7889 0.7943 0.8149 1.0323 1.0548 1.0599
#> speed=~x7        1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> speed=~x8        1.1800 0.1645 100 0.8443 0.8692 0.9165 1.5372 1.6276 1.6632
#> speed=~x9        1.0815 0.1442 100 0.7945 0.8077 0.8275 1.3836 1.4658 1.5199
#> x1~~x1           0.5491 0.1232 100 0.2103 0.2113 0.2671 0.7411 0.8158 0.8158
#> x2~~x2           1.1338 0.0981 100 0.8187 0.8669 0.9795 1.3254 1.3461 1.3574
#> x3~~x3           0.8443 0.0986 100 0.5445 0.6078 0.6876 1.0371 1.0821 1.0901
#> x4~~x4           0.3712 0.0495 100 0.2357 0.2564 0.2900 0.4625 0.5028 0.5293
#> x5~~x5           0.4463 0.0568 100 0.3177 0.3254 0.3451 0.5712 0.5920 0.6059
#> x6~~x6           0.3562 0.0415 100 0.2442 0.2546 0.2717 0.4303 0.4440 0.4444
#> x7~~x7           0.7994 0.0806 100 0.6236 0.6321 0.6587 0.9455 1.0475 1.0698
#> x8~~x8           0.4877 0.0755 100 0.2858 0.3163 0.3570 0.6501 0.6735 0.6880
#> x9~~x9           0.5661 0.0652 100 0.3270 0.3676 0.4368 0.6843 0.7001 0.7114
#> visual~~visual   0.8093 0.1566 100 0.4745 0.4779 0.5262 1.1124 1.1807 1.2061
#> textual~~textual 0.9795 0.1147 100 0.6911 0.6990 0.7551 1.1753 1.2905 1.3271
#> speed~~speed     0.3837 0.0784 100 0.1327 0.1641 0.2342 0.5296 0.5426 0.5508
#> visual~~textual  0.4082 0.0724 100 0.2519 0.2534 0.2836 0.5375 0.5721 0.5797
#> visual~~speed    0.2622 0.0528 100 0.1368 0.1480 0.1741 0.3683 0.3828 0.3861
#> textual~~speed   0.1735 0.0500 100 0.0184 0.0546 0.0983 0.2720 0.2897 0.3006
#> Standardized Monte Carlo Confidence Intervals
#>                     est     se   R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> visual=~x1       0.7719 0.0531 100 0.6020 0.6255 0.6598 0.8533 0.8972 0.8987
#> visual=~x2       0.4236 0.0631 100 0.2491 0.2594 0.2994 0.5510 0.5822 0.5886
#> visual=~x3       0.5811 0.0575 100 0.4153 0.4238 0.4443 0.6778 0.6971 0.6995
#> textual=~x4      0.8516 0.0221 100 0.8005 0.8025 0.8053 0.8860 0.9059 0.9116
#> textual=~x5      0.8551 0.0260 100 0.7703 0.7864 0.8101 0.8966 0.9113 0.9123
#> textual=~x6      0.8380 0.0234 100 0.7657 0.7728 0.7954 0.8765 0.8842 0.8843
#> speed=~x7        0.5695 0.0558 100 0.3507 0.3715 0.4378 0.6458 0.6538 0.6563
#> speed=~x8        0.7230 0.0492 100 0.5553 0.5680 0.5924 0.7997 0.8064 0.8070
#> speed=~x9        0.6650 0.0501 100 0.5310 0.5359 0.5495 0.7544 0.7607 0.7614
#> x1~~x1           0.4042 0.0807 100 0.1924 0.1951 0.2718 0.5647 0.6080 0.6375
#> x2~~x2           0.8206 0.0535 100 0.6535 0.6610 0.6964 0.9103 0.9326 0.9379
#> x3~~x3           0.6623 0.0652 100 0.5107 0.5141 0.5406 0.8026 0.8203 0.8275
#> x4~~x4           0.2748 0.0375 100 0.1689 0.1794 0.2150 0.3515 0.3559 0.3592
#> x5~~x5           0.2689 0.0443 100 0.1677 0.1696 0.1961 0.3438 0.3813 0.4066
#> x6~~x6           0.2977 0.0390 100 0.2180 0.2182 0.2318 0.3674 0.4028 0.4138
#> x7~~x7           0.6757 0.0602 100 0.5692 0.5725 0.5830 0.8075 0.8614 0.8769
#> x8~~x8           0.4772 0.0688 100 0.3487 0.3497 0.3604 0.6490 0.6772 0.6916
#> x9~~x9           0.5578 0.0652 100 0.4203 0.4214 0.4309 0.6980 0.7128 0.7180
#> visual~~visual   1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> textual~~textual 1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> speed~~speed     1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> visual~~textual  0.4585 0.0640 100 0.2842 0.2907 0.3145 0.5778 0.6175 0.6304
```

```
#> visual~~speed    0.4705 0.0776 100 0.3007 0.3064 0.3201 0.6360 0.6839 0.6974
#> textual~~speed   0.2830 0.0630 100 0.1363 0.1429 0.1595 0.3828 0.3890 0.3920
#> Standardized Monte Carlo Confidence Intervals
#>                     est     se   R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> visual=~x1       0.7719 0.0610 100 0.6069 0.6120 0.6421 0.8906 0.9209 0.9221
#> visual=~x2       0.4236 0.0625 100 0.2157 0.2200 0.2806 0.5455 0.5680 0.5685
#> visual=~x3       0.5811 0.0625 100 0.3004 0.3442 0.4381 0.6517 0.6662 0.6701
#> textual=~x4      0.8516 0.0234 100 0.7885 0.7897 0.8025 0.8874 0.9043 0.9130
#> textual=~x5      0.8551 0.0218 100 0.7918 0.7929 0.8018 0.8856 0.8993 0.9045
#> textual=~x6      0.8380 0.0234 100 0.7748 0.7766 0.7971 0.8852 0.8904 0.8915
#> speed=~x7        0.5695 0.0533 100 0.3529 0.3969 0.4532 0.6496 0.6621 0.6674
#> speed=~x8        0.7230 0.0511 100 0.5945 0.5964 0.6124 0.8149 0.8171 0.8181
#> speed=~x9        0.6650 0.0498 100 0.5282 0.5508 0.5807 0.7518 0.7936 0.8123
#> x1~~x1           0.4042 0.0938 100 0.1497 0.1519 0.2068 0.5875 0.6254 0.6317
#> x2~~x2           0.8206 0.0514 100 0.6768 0.6773 0.7025 0.9209 0.9516 0.9535
#> x3~~x3           0.6623 0.0665 100 0.5509 0.5562 0.5752 0.8072 0.8791 0.9093
#> x4~~x4           0.2748 0.0395 100 0.1663 0.1822 0.2124 0.3560 0.3763 0.3783
#> x5~~x5           0.2689 0.0370 100 0.1819 0.1911 0.2157 0.3572 0.3713 0.3730
#> x6~~x6           0.2977 0.0392 100 0.2052 0.2072 0.2165 0.3646 0.3969 0.3997
#> x7~~x7           0.6757 0.0579 100 0.5546 0.5616 0.5780 0.7946 0.8401 0.8750
#> x8~~x8           0.4772 0.0730 100 0.3307 0.3323 0.3359 0.6250 0.6443 0.6466
#> x9~~x9           0.5578 0.0665 100 0.3401 0.3697 0.4348 0.6628 0.6960 0.7209
#> visual~~visual   1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> textual~~textual 1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> speed~~speed     1.0000 0.0000 100 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
#> visual~~textual  0.4585 0.0566 100 0.3183 0.3257 0.3461 0.5577 0.5782 0.5803
#> visual~~speed    0.4705 0.0762 100 0.2804 0.2901 0.3417 0.6249 0.6431 0.6441
#> textual~~speed   0.2830 0.0741 100 0.0358 0.0984 0.1786 0.4111 0.4650 0.4973
#> Test passed with 1 success.
```

#> test-semmcci-npd

```
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-semmcci-zzz-coverage

```
#> Test passed with 1 success.
```

#> test-semmcci-zzz-heywood

```
#> lavaan 0.6-21 ended normally after 18 iterations
#>
#>   Estimator                                         ML
#>   Optimization method                           NLMINB
```

```
#>   Number of model parameters                         8
#>
#>   Number of observations                            50
#>
#> Model Test User Model:
#>
#>   Test statistic                                 0.000
#>   Degrees of freedom                                 2
#>   P-value (Chi-square)                           1.000
#>
#> Parameter Estimates:
#>
#>   Standard errors                             Standard
#>   Information                                 Expected
#>   Information saturated (h1) model          Structured
#>
#> Latent Variables:
#>                    Estimate  Std.Err  z-value  P(>|z|)
#>   eta1 =~
#>     y1                1.000
#>     y2                0.991    0.152    6.524    0.000
#>     y3                0.901    0.138    6.537    0.000
#>     y4                1.399    0.143    9.814    0.000
#>
#> Variances:
#>                    Estimate  Std.Err  z-value  P(>|z|)
#>    .y1                0.000    0.055    0.000    1.000
#>    .y2                0.980    0.203    4.822    0.000
#>    .y3                0.807    0.167    4.820    0.000
#>    .y4                0.696    0.176    3.960    0.000
#>     eta1              0.980    0.204    4.815    0.000
#>
#> Standardized Monte Carlo Confidence Intervals
#>                est se R 0.05% 0.5% 2.5% 97.5% 99.5% 99.95%
#> eta1=~y1   1.0000 NA 0    NA   NA   NA    NA    NA     NA
#> eta1=~y2   0.7039 NA 0    NA   NA   NA    NA    NA     NA
#> eta1=~y3   0.7047 NA 0    NA   NA   NA    NA    NA     NA
#> eta1=~y4   0.8566 NA 0    NA   NA   NA    NA    NA     NA
#> y1~~y1     0.0000 NA 0    NA   NA   NA    NA    NA     NA
#> y2~~y2     0.5045 NA 0    NA   NA   NA    NA    NA     NA
#> y3~~y3     0.5034 NA 0    NA   NA   NA    NA    NA     NA
#> y4~~y4     0.2662 NA 0    NA   NA   NA    NA    NA     NA
#> eta1~~eta1 1.0000 NA 0    NA   NA   NA    NA    NA     NA
#> Test passed with 1 success.

#> test-semmcci
```

```
#> Test passed with 1 success.
#> [[1]]
#> [[1]][[1]]
#> [[1]][[1]]$value
#> [[1]][[1]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[1]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[2]]
#> [[1]][[2]]$value
#> [[1]][[2]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
```

```
#>
#>
#> [[1]][[5]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[6]]
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[7]]
#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[8]]
#> [[1]][[8]]$value
#> [[1]][[8]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[8]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[9]]
#> [[1]][[9]]$value
#> [[1]][[9]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[9]]$visible
#> [1] TRUE
#>
#>
```

```
#> [[1]][[10]]
#> [[1]][[10]]$value
#> [[1]][[10]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[10]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[11]]
#> [[1]][[11]]$value
#> [[1]][[11]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[11]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[12]]
#> [[1]][[12]]$value
#> [[1]][[12]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[12]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[13]]
#> [[1]][[13]]$value
#> [[1]][[13]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[13]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[14]]
#> [[1]][[14]]$value
#> [[1]][[14]]$value[[1]]
#> [1] TRUE
#>
```

```
#>
#> [[1]][[14]]$visible
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#>
#>
#> [[1]][[15]]
#> [[1]][[15]]$value
#> [[1]][[15]]$value[[1]]
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#>
#> [[1]][[15]]$visible
#> [1] TRUE
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#>
#> [[1]][[16]]
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#> [[1]][[16]]$visible
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#>
#> [[1]][[17]]
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#> [1] TRUE
#>
#>
#> [[1]][[17]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[18]]
#> [[1]][[18]]$value
#> [[1]][[18]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[18]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[19]]
```

```
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#> [[1]][[19]]$value[[1]]
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#>
#>
#> [[1]][[19]]$visible
#> [1] TRUE
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#>
#> [[1]][[20]]
#> [[1]][[20]]$value
#> [[1]][[20]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[20]]$visible
#> [1] TRUE
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#>
#> [[1]][[21]]
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#> [[1]][[21]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[21]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[22]]
#> [[1]][[22]]$value
#> [[1]][[22]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[22]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[23]]
#> [[1]][[23]]$value
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#>
#>
```

```
#> [[1]][[23]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[24]]
#> [[1]][[24]]$value
#> [[1]][[24]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[24]]$visible
#> [1] TRUE
```

Environment

```
ls()

#> [1] "root"
```

Class

```
#> [[1]]
#> [1] "root_criterion"
```

References

Pesigan, I. J. A., & Cheung, S. F. (2024). Monte Carlo confidence intervals for the indirect effect with missing data. *Behavior Research Methods*, *56*(3), 1678–1696. https://doi.org/10.3758/s13428-023-02114-4

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/